CSEC Add Maths

Paper 2

June 2025

Questions

Kerwin Springer

Best Online Lessons in the Caribbean
WhatsApp **+1868-310-1306** for more information

SECTION I

ALGEBRA, SEQUENCES AND SERIES

ALL working must be clearly shown.

1. (a) Solve the equation $3^{2x+1} = 21$, giving your answer in exact form. [4]

$$3^{2x+1} = 21$$

$$3^{2x+1} = 7 \times 3^1$$

$$\frac{3^{2x+1}}{3^1} = 7$$

$$3^{2x+1-1} = 7$$

$$3^{2x} = 7$$

Taking ln on both sides:

$$\ln 3^{2x} = \ln 7$$

$$2x \ln 3 = \ln 7$$

$$2x = \frac{\ln 7}{\ln 3}$$

$$x = \frac{\ln 7}{2\ln 3}$$

Kerwin Springer

(b) Determine the values of x and y which satisfy the equations

$$y + 3x = 2 \quad \text{and} \quad y + 4 = x(5 - 2x). \qquad [6]$$

$y + 3x = 2$ —— Equation 1

$y + 4 = x(5 - 2x)$ —— Equation 2

Rearranging Equation 1 to make y the subject of the formula:

$y = 2 - 3x$ —— Equation 3

Substituting Equation 3 into Equation 2:

$$(2 - 3x) + 4 = x(5 - 2x)$$

$$2 - 3x + 4 = 5x - 2x^2$$

$$-3x + 6 = 5x - 2x^2$$

$$2x^2 - 5x - 3x + 6 = 0$$

$$2x^2 - 8x + 6 = 0$$

Dividing throughout by 2:

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

Either

$x - 1 = 0$ or $x - 3 = 0$

$x = 1$ or $x = 3$

Kerwin Springer

STUDENT HUB

Kerwin Springer

Best Online Lessons in the Caribbean
WhatsApp **+1868-310-1306** for more information

When $x = 1$,

$y = 2 - 3(1)$

$y = 2 - 3$

$y = -1$

When $x = 3$,

$y = 2 - 3(3)$

$y = 2 - 9$

$y = -7$

$\therefore$ The values of x and y which satisfy the equations are:

$x = 1, y = -1$ and $x = 3, y = -7$.

Kerwin Springer

(c) (i) Determine the solution set for the inequality $\frac{(3x+2)}{(x-1)} \leq 0$, giving your answer in set builder notation. [4]

$$\frac{(3x+2)}{(x-1)} \leq 0$$

Multiplying by $(x-1)^2$:

$$\frac{(3x+2)}{(x-1)} \times (x-1)^2 \leq 0 \times (x-1)^2$$

$$(3x+2)(x-1) \leq 0$$

Roots occur when

$3x + 2 = 0$ and $x - 1 = 0$

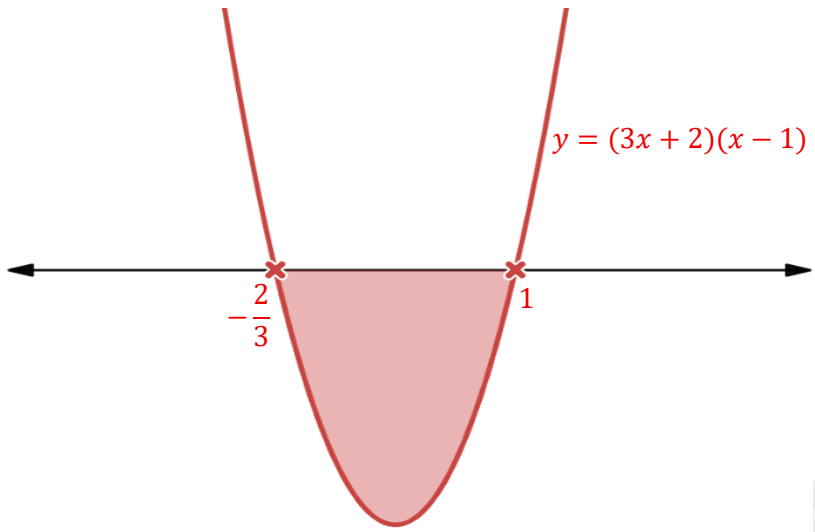
$3x = -2$ and $x = 1$

$x = -\frac{2}{3}$

- The coefficient of x^2 is positive therefore, this graph will have a <u>minimum curve</u>.
- Roots: $\left(-\frac{2}{3}, 0\right)$ and $(1, 0)$
- Since $(3x+2)(x-1) \leq 0$, this is the region where $y \leq 0$.

A sketch of the curve $y = (3x+2)(x-1)$ looks like:

Kerwin Springer



$\therefore$ The solution set is $\{x\colon\ -\frac{2}{3} \leq x < 1\}$

Kerwin Springer

(ii) State how the solution of $(3x+2)(x-1) \leq 0$ would differ from the solution of $\frac{(3x+2)}{(x-1)} \leq 0$ [1]

The solution of $(3x+2)(x-1) \leq 0$ is $\{x: -\frac{2}{3} \leq x \leq 1\}$.

The solution of $\frac{(3x+2)}{(x-1)} \leq 0$ is $\{x: -\frac{2}{3} \leq x < 1\}$.

The difference between the two solutions is the exclusion of 1 in the solution for $\frac{(3x+2)}{(x-1)} \leq 0$.

1 is excluded because it would make the function undefined.

Total: 15 marks

Kerwin Springer

2. (a) (i) The relationship between two variables, x and y, is $10^{2x} \times 10^{y} = 3$. Express the relationship in the form $y = \log b + ax$, where $a, b \in R$. [4]

$$10^{2x} \times 10^{y} = 3$$

$$10^{2x+y} = 3$$

Taking log_{10} on both sides:

$$log_{10}\,(10^{2x+y}) = log_{10}\,3$$

$$log_{10}\,10^{2x+y} = log_{10}\,3$$

$$(2x + y)log_{10} = log_{10}\,3$$

$$2x + y = \frac{log_{10}\,3}{log_{10}\,10}$$

$$2x + y = \frac{log_{10}\,3}{1}$$

$$2x + y = log_{10}\,3$$

$$y = log_{10}\,3 - 2x$$

This is of the form $y = \log b + ax$ where $a = -2$ and $b = 3$.

Kerwin Springer

(ii) Hence, state the value of the y-intercept of a graph of y versus x. [1]

$y = log_{10} 3 - 2x$

Rearranging:

$y = -2x + log_{10} 3$

This is of the form $y = mx + c$ where $m = -2$ and $c = log_{10} 3$.

Therefore, the value of the y-intercept of a graph is $c = log_{10} 3$.

Kerwin Springer

(b) Determine whether the series $2 + 6 + 18 + 54 + 162 \ldots$ is arithmetic or geometric. Justify your response. [2]

An arithmetic series has a common difference between the terms.

From the series given, there is no common difference, d as shown below:

$T_2 - T_1$: $6 - 2 = 4$

$T_3 - T_2$: $18 - 6 = 12$

$T_4 - T_3$: $54 - 18 = 36$

$T_5 - T_4$: $162 - 54 = 108$

Hence, it is not an Arithmetic Progression.

A geometric series has a common ratio between the terms.

From the series given, there is a common ratio, r as shown below:

$\frac{T_2}{T_1}$: $\frac{6}{2} = 3$

$\frac{T_3}{T_2}$: $\frac{18}{6} = 3$

$\frac{T_4}{T_3}$: $\frac{54}{18} = 3$

$\frac{T_5}{T_4}$: $\frac{162}{54} = 3$

Since the sequence has a common ratio, it is geometric in nature.

Kerwin Springer

(c) In a geometric progression, the 3rd term is 25 and the sum of the 1st and 2nd terms is 500. Find the common ratio where $r > 0$. [4]

In a geometric progression, $T_n = ar^{n-1}$, where $T_n = n^{th}$ term, $a = 1^{st}$ term, $r =$ common ratio.

From the question, the following information is given:

"The 3rd term is 25."

$$T_3 = 25$$

$$ar^2 = 25$$

$$a = \frac{25}{r^2} \quad \text{Equation 1}$$

"The sum of the 1st and 2nd terms is 500"

$$T_1 + T_2 = 500$$

$$a + ar = 500 \quad \text{Equation 2}$$

Substituting Equation 1 into Equation 2:

$$\left(\frac{25}{r^2}\right) + \left(\frac{25}{r^2}\right)r = 500$$

$$\frac{25}{r^2} + \frac{25}{r} = 500$$

Kerwin Springer

$$\frac{25 + 25r}{r^2} = 500$$

$$25 + 25r = 500r^2$$

$$0 = 500r^2 - 25r - 25$$

$$500r^2 - 25r - 25 = 0$$

Dividing throughout by 25:

$$20r^2 - r - 1 = 0$$

$$(5r + 1)(4r - 1) = 0$$

Either

$5r + 1 = 0$ or $4r - 1 = 0$

$5r = -1$ or $4r = 1$

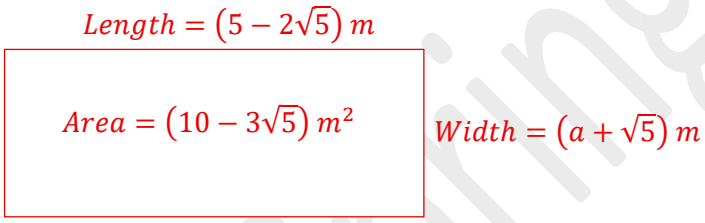
$r = -\frac{1}{5}$ or $r = \frac{1}{4}$

Since $r > 0$, then $r = \frac{1}{4}$ only.

Kerwin Springer

(d) A kitchen garden has an area, A, $(10 - 3\sqrt{5})\ m^2$. The length of the garden is $(5 - 2\sqrt{5})\ m$. Determine the value of a, if the width of the garden, in metres, is in the form $(a + \sqrt{5})$, where a is a real number. [4]

Assuming the kitchen garden is rectangular, a sketch is shown below.



<u>Method 1</u>

$$Area = length \times width$$

$$Width = \frac{Area}{length}$$

$$= \frac{(10 - 3\sqrt{5})}{(5 - 2\sqrt{5})}$$

Rationalizing the denominator:

$$Width = \frac{(10 - 3\sqrt{5})}{(5 - 2\sqrt{5})} \times \frac{(5 + 2\sqrt{5})}{(5 + 2\sqrt{5})}$$

$$= \frac{50 + 20\sqrt{5} - 15\sqrt{5} - 6(5)}{25 + 10\sqrt{5} - 10\sqrt{5} - 4(5)}$$

Kerwin Springer

$$= \frac{20 + 5\sqrt{5}}{5}$$

$$= \frac{5(4 + \sqrt{5})}{5}$$

$$= 4 + \sqrt{5}$$

which is of the form $(a + \sqrt{5})$

Therefore, $a = 4$.

<u>Method 2</u>

$Area = length \times width$

$(10 - 3\sqrt{5}) = (5 - 2\sqrt{5}) \times (a + \sqrt{5})$

$10 - 3\sqrt{5} = 5a + 5\sqrt{5} - 2a\sqrt{5} - 2(5)$

$10 - 3\sqrt{5} = 5a - 10 + 5\sqrt{5} - 2a\sqrt{5}$

$10 - 3\sqrt{5} = (5a - 10) + (5 - 2a)\sqrt{5}$

Equating coefficients:

$10 = 5a - 10$

$10 + 10 = 5a$

$20 = 5a$

$\frac{20}{5} = a$

Kerwin Springer

$a = 4$

OR

$-3 = 5 - 2a$

$-3 - 5 = -2a$

$-8 = -2a$

$\frac{-8}{-2} = a$

$a = 4$

Therefore, $a = 4$.

Kerwin Springer

Total: 15 marks

SECTION II

COORDINATE GEOMETRY, VECTORS AND TRIGONOMETRY

ALL working must be clearly shown.

3. (a) (i) Using the grid on page 11, draw the graphs of $\sin 2x$ and $\cos 2x$, for

$0 \leq x \leq \pi$ [2]

Table of values for $y = sin\ 2x$

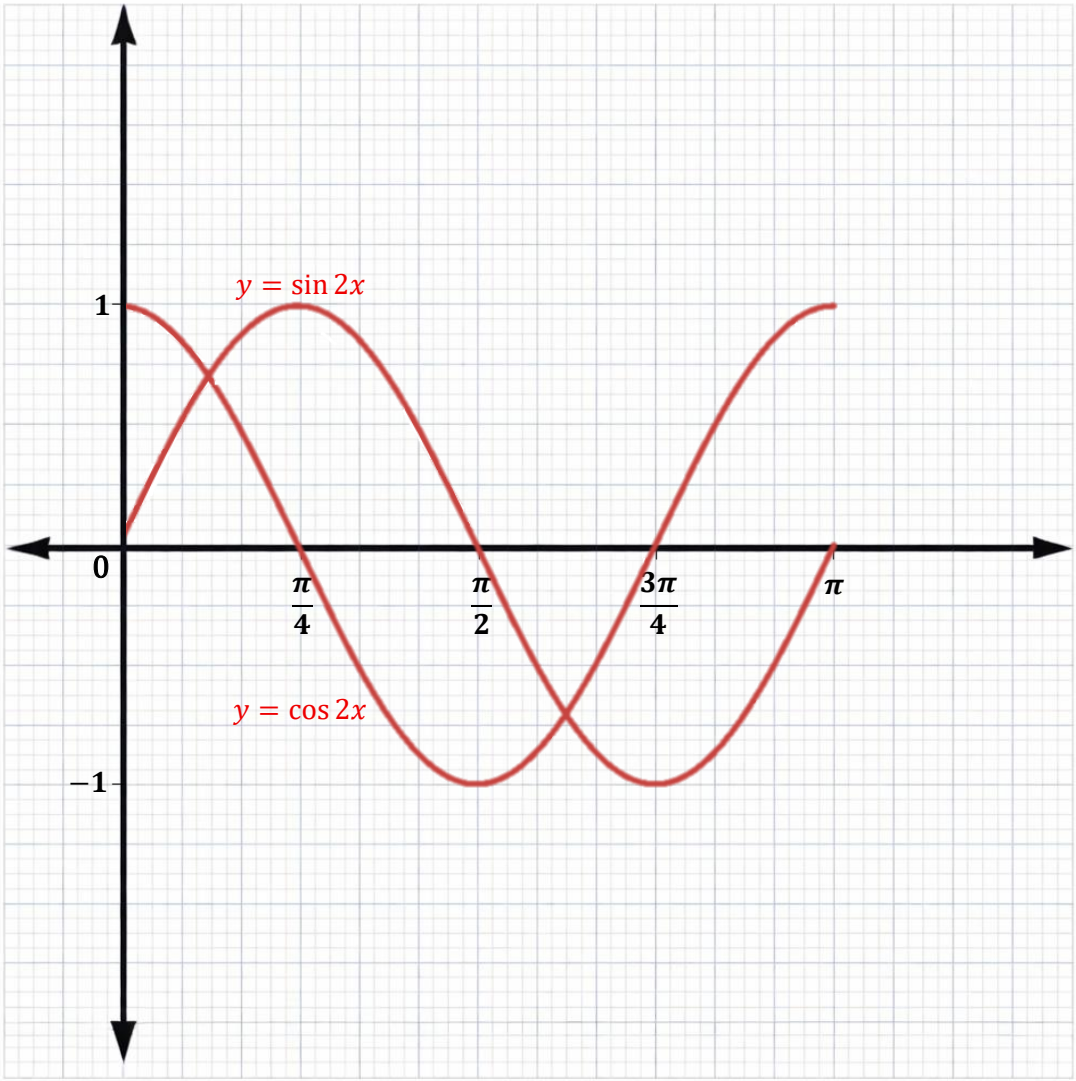
x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$\sin 2x$	0	1	0	-1	0

Table of values for $y = cos\ 2x$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$\cos 2x$	1	0	-1	0	1

The graphs of both $\sin 2x$ and $\cos 2x$ can be seen on the next page.

Kerwin Springer


$y = \sin 2x$
$y = \cos 2x$
1
−1
0
$\frac{\pi}{4}$
$\frac{\pi}{2}$
$\frac{3\pi}{4}$
π

(ii) Hence or otherwise, solve the equation $\sin 2x = \cos 2x$ for $0 \leq x \leq \pi$. [4]

. $\sin 2x = \cos 2x$

$$\frac{\sin 2x}{\cos 2x} = \frac{\cos 2x}{\cos 2x}$$

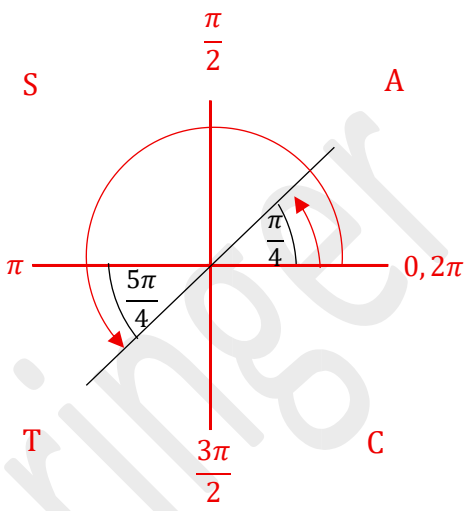
$$\tan 2x = 1$$

$$2x = \tan^{-1}(1)$$

$$2x = \frac{\pi}{4}, \left(\pi + \frac{\pi}{4}\right)$$

$$2x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$x = \frac{\pi}{8}, \frac{5\pi}{8}$$



$\therefore x = \frac{\pi}{8}, \frac{5\pi}{8}$ for $0 \leq x \leq \pi$.

Kerwin Springer

(b) (i) Using an appropriate double-angle formula, derive the identity for $\sin 2x$.

[2]

Recall:- Compound angle formula

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

When $B = A$,

$\sin(A+A) = \sin A \cos A + \cos A \sin A$

$\sin 2A = 2 \sin A \cos A$

$\therefore \quad \sin 2x = 2 \sin x \cos x$

(ii) Hence, prove the identity $\frac{1+\sin 2x}{\sin x + \cos x} = \sin x + \cos x$. [4]

L.H.S.

$$\frac{1 + \sin 2x}{\sin x + \cos x} = \frac{1 + \sin 2x}{\sin x + \cos x} \times \frac{\sin x + \cos x}{\sin x + \cos x}$$

$$= \frac{(1 + \sin 2x)(\sin x + \cos x)}{(\sin x + \cos x)(\sin x + \cos x)}$$

$$= \frac{(1 + \sin 2x)(\sin x + \cos x)}{\sin^2 x + \cos^2 x + 2 \sin x \cos x}$$

Recall: $\sin^2 x + \cos^2 x = 1$ and $\sin 2x = 2 \sin x \cos x$

$$= \frac{\cancel{(1 + \sin 2x)}(\sin x + \cos x)}{\cancel{(1 + \sin 2x)}}$$

$$= \sin x + \cos x$$

$$= \text{R.H.S.}$$

Q.E.D.

Kervin Springer

(c) The position vectors of two points, A and B, whose coordinates are $(2, 3)$ and $(-3, 2)$ respectively are relative to an origin, O.

(i) Express the unit vector $\overrightarrow{AB}$ in terms of $x\boldsymbol{i} + y\boldsymbol{i}$. [4]

Since $A = (2, 3)$ then $\overrightarrow{OA} = \binom{2}{3}$

$= 2\boldsymbol{i} + 3\boldsymbol{j}$

Since $B = (-3, 2)$ then $\overrightarrow{OB} = \binom{-3}{2}$

$= -3\boldsymbol{i} + 2\boldsymbol{j}$

Using the triangle law,

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{0A}$

$= \binom{-3}{2} - \binom{2}{3}$

$= \binom{-5}{-1}$

$= -5\boldsymbol{i} - \boldsymbol{j}$

$|AB| = \sqrt{(-5)^2 + (-1)^2}$

$= \sqrt{25 + 1}$

$= \sqrt{26}$

Substituting $\overrightarrow{AB} = -5\boldsymbol{i} - \boldsymbol{j}$ and $|AB| = \sqrt{26}$ into

$$Unit\ Vector\ of\ AB = \frac{\overrightarrow{AB}}{|AB|}$$

$$= \frac{-5\boldsymbol{i} - \boldsymbol{j}}{\sqrt{26}}$$

$$= \frac{1}{\sqrt{26}}(-5\boldsymbol{i} - \boldsymbol{j}) \quad or \quad -\frac{5}{\sqrt{26}}\boldsymbol{i} - \frac{1}{\sqrt{26}}\boldsymbol{j}$$

Kerwin Springer

(ii) Find the angle $A\hat{O}B$, in degrees. [3]

$\overrightarrow{OA} = \binom{2}{3}$ $\overrightarrow{OB} = \binom{-3}{2}$

Using the dot product

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = (2 \times -3) + (3 \times 2)$$
$$= -6 + 6$$
$$= 0$$

$$|OA| = \sqrt{(2)^2 + (3)^2}$$
$$= \sqrt{4 + 9}$$
$$= \sqrt{13}$$

$$|OB| = \sqrt{(-3)^2 + (2)^2}$$
$$= \sqrt{9 + 3}$$
$$= \sqrt{13}$$

$$\cos\theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|OA||OB|}$$
$$\cos\theta = \frac{0}{\sqrt{13}\sqrt{13}}$$
$$\cos\theta = 0$$
$$\theta = \cos^{-1} 0$$
$$\theta = 90°$$

Therefore, angle $AOB = 90°$.

Kerwin Springer

(iii) Hence, state the relationship between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [1]

Since angle $AOB = 90°$, then the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are perpendicular to each other.

Total: 20 marks

Kerwin Springer

SECTION III

INTRODUCTORY CALCULUS

ALL working must be clearly shown.

4. (a) Differentiate the following expression with respect to x, simplifying your answer.

$$y = \sin 4x + \frac{3}{x^4}$$

[3]

$y = \sin 4x + \frac{3}{x^4}$

$y = \sin 4x + 3x^{-4}$

$\frac{dy}{dx} = 4\cos 4x + (-4)(3)x^{-4-1}$

$= 4\cos 4x - 12\,x^{-5}$

$= 4\cos 4x - \frac{12}{x^5}$

Kerwin Springer

(b) (i) Determine the coordinates of all the stationary points of the curve

$$y = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x - \frac{1}{2}$$

[4]

$$y = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x - \frac{1}{2}$$

$$= \frac{1}{3}x^3 - \frac{5}{2}x^2 + 4x^1 - \frac{1}{2}x^0$$

$$\frac{dy}{dx} = 3\left(\frac{1}{3}x^{3-1}\right) - 2\left(\frac{5}{2}x^{2-1}\right) + 1(4x^{1-1}) - 0\left(\frac{1}{2}x^{0-1}\right)$$

$$= x^2 - 5x + 4$$

At stationary points, $\frac{dy}{dx} = 0$.

$x^2 - 5x + 4 = 0$

$(x-1)(x-4) = 0$

Either

$x - 1 = 0$ or $x - 4 = 0$

$x = 1$ or $x = 4$

When $x = 1$,

$$y = \frac{1}{3}(1)^3 - \frac{5}{2}(1)^2 + 4(1) - \frac{1}{2}$$

$$y = \frac{1}{3} - \frac{5}{2} + 4 - \frac{1}{2}$$

$$y = \frac{4}{3}$$

Kerwin Springer

When $x = 4$,

$$y = \frac{1}{3}(4)^3 - \frac{5}{2}(4)^2 + 4(4) - \frac{1}{2}$$

$$y = \frac{64}{3} - 40 + 16 - \frac{1}{2}$$

$$y = -\frac{19}{6}$$

$\therefore$ The stationary points are: $\left(1, \frac{4}{3}\right)$ and $\left(4, -\frac{19}{6}\right)$

Kerwin Springer

(ii) Hence, determine the nature of EACH of the stationary points found in (b) (i). [3]

$$\frac{dy}{dx} = x^2 - 5x + 4$$

$$= x^2 - 5x^1 + 4x^0$$

$$\frac{d^2y}{dx^2} = 2(x^{2-1}) - 1(5x^{1-1}) + 0(4x^{0-1})$$

$$= 2x - 5 + 0$$

$$= 2x - 5$$

When $x = 1$,

$$\frac{d^2y}{dx^2} = 2(1) - 5$$

$$= 2 - 5$$

$$= -3$$

Since $\frac{d^2y}{dx^2} < 0$, then the stationary point $\left(1, \frac{4}{3}\right)$ is a maximum point.

When $x = 4$,

$$\frac{d^2y}{dx^2} = 2(4) - 5$$

$$= 8 - 5$$

$$= 3$$

Since $\frac{d^2y}{dx^2} > 0$, then the stationary point $\left(4, -\frac{19}{6}\right)$ is a minimum point.

Kerwin Springer

(c) The length of each edge of a cube is increasing at a rate of 0.25 $cm\ s^{-1}$.

Calculate the rate of change of the volume of the cube at the instant when the length is 8 cm. [5]

From the question, the following is given:

$$\frac{dx}{dt} = 0.25\ cm\ s^{-1}$$

$$Volume,\ \ v = x \times x \times x\ cm^3$$

$$= x^3\ cm^3$$

$$\frac{dv}{dx} = 3x^2$$

When $x = 8$,

$$\frac{dv}{dx} = 3(8)^2$$

$$= 192$$

Required to find: $\frac{dv}{dt}$

Using the chain rule,

$$\frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{dt}$$

$$= 192 \times 0.25$$

$$= 48\ cm^3\ s^{-1}$$

$\therefore$ The rate of change of the volume is $\frac{dv}{dt} = 48\ cm^3\ s^{-1}$.

Kerwin Springer

Total: 15 marks

5. (a) Determine

(i) $\int \sqrt{2x+1}\ dx$ [3]

$$\int \sqrt{2x+1}\ dx$$

$$= \int (2x+1)^{\frac{1}{2}}\ dx$$

$$= \frac{(2x+1)^{\frac{1}{2}+1}}{2\left(\frac{1}{2}+1\right)} + c$$

$$= \frac{(2x+1)^{\frac{3}{2}}}{3} + c$$

$$= \frac{1}{3}(2x+1)^{\frac{3}{2}} + c$$

$$\therefore \int \sqrt{2x+1}\ dx = \frac{1}{3}(2x+1)^{\frac{3}{2}} + c$$

Kerwin Springer

(ii) $\int \sin 2x + 5\cos x \ dx$ [3]

$$\int \sin 2x + 5\cos x \ dx$$

$$= \int \sin 2x + \int 5\cos x \, dx$$

Recall: $\int \sin ax \ dx = \frac{-\cos ax}{a} + c$

$$= \frac{-\cos 2x}{2} + 5\sin x + c$$

$$= -\frac{1}{2}\cos 2x \ + 5\sin x + c$$

Kerwin Springer

(b) Figure 1 shows the region enclosed between the x-$axis$, the y-$axis$, the curve $y = x^2 + 2$ and the line $x = 3$.

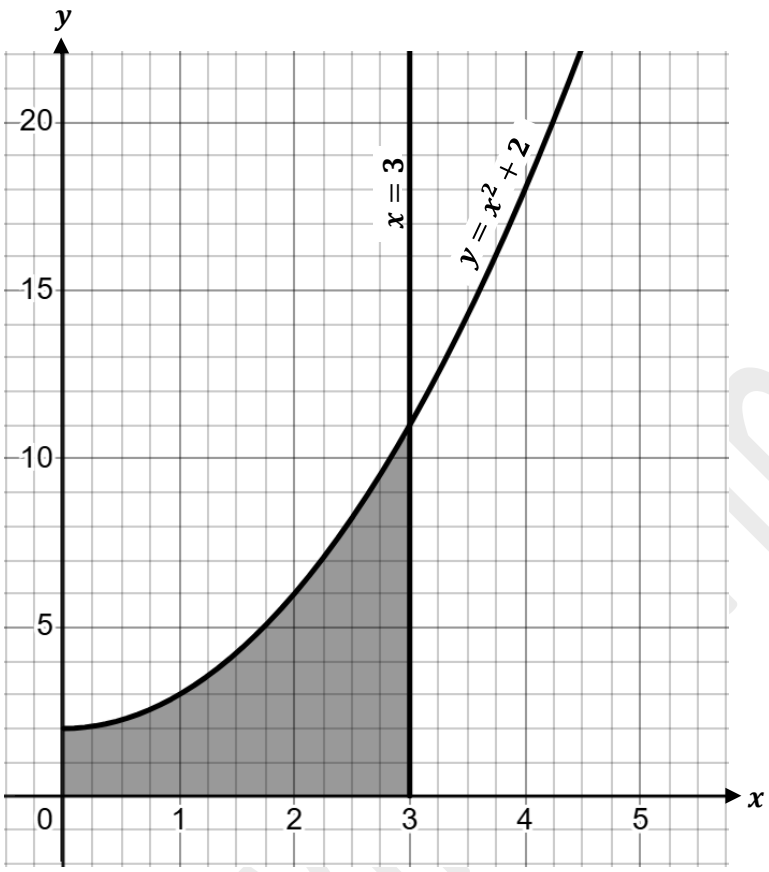


Figure 1. Region bounded by a curve, a line, the x-axis and y-axis.

Calculate the area of the shaded region. [4]

From the question, $y = x^2 + 2$, $\quad a = 0$, $\quad b = 3$.

$$Area\ of\ shaded\ region,\ \ A = \int_a^b y\ dx$$

$$= \int_0^3 x^2 + 2\ dx$$

$$= \left[\frac{x^3}{3} + 2x\right]_0^3$$

$$= \left[\frac{(3)^3}{3} + 2(3)\right] - \left[\frac{(0)^3}{3} + 2(0)\right]$$

$$= 15\ units^2$$

Kerwin Springer

(c) Figure 2 shows the shaded region enclosed between the x-axis, the y-axis and the curve $y = 4 - x^2$.

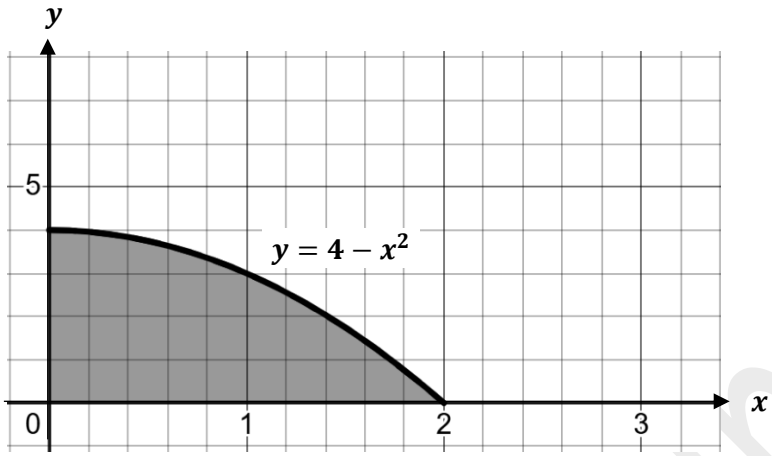


Figure 2. Region bounded by a curve, the x-axis, and y-axis.

If the region is rotated about the x-axis through one complete revolution, calculate the volume of the solid formed. [5]

From the question, $y = 4 - x^2$, $\quad a = 0, \quad b = 2.$

$$Volume,\ \ V = \pi \int_a^b y^2\ dx$$

$$= \pi \int_0^2 (4 - x^2)^2\ dx$$

$$= \pi \int_0^2 (16 - 8x^2 + x^4)\ dx$$

$$= \pi \int_0^2 (16x^0 - 8x^2 + x^4)\ dx$$

$$= \pi \left[\frac{16x^{0+1}}{1} - \frac{8x^{2+1}}{3} + \frac{x^{4+1}}{5}\right]_0^2$$

Kerwin Springer

$$= \pi\left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_0^2$$

$$= \pi\left[\left(16(2) - \frac{8(2)^3}{3} + \frac{(2)^5}{5}\right) - \left(16(0) - \frac{8(0)^3}{3} + \frac{(0)^5}{5}\right)\right]$$

$$= \pi\left[\left(32 - \frac{64}{3} + \frac{32}{5}\right) - 0\right]$$

$$= \pi\left[\frac{256}{15}\right]$$

$$= \frac{256}{15}\pi \quad units^3$$

Total: 15 marks

Kerwin Springer

SECTION IV

PROBABILITY AND STATISTICS

ALL working must be clearly shown.

6. (a) If $P(A) = 0.8$, $P(B) = 0.5$ and $P(A \cup B) = 0.9$, calculate $P(B|A)$ [5]

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.8 + 0.5 - 0.9$$

$$= 0.4$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$= \frac{0.4}{0.8}$$

$$= 0.5$$

$$\therefore P(B|A) = 0.5$$

Kerwin Springer

(b) A fair, blue dice has each of the numbers 1, 3, 5, 7, 9 and 11 on its faces. A second, fair, red dice has each of the numbers 2, 4, 6, 8, 10 and 12 on its faces. The dice are rolled simultaneously and the sum of the numbers on their faces is recorded.

(i) Complete the possibility space table below by inserting the sum of the two dice on each roll. The first sum and the last sum have been inserted for you. [3]

		Red Dice					
	+	**2**	**4**	**6**	**8**	**10**	**12**
Blue Dice	**1**	3	5	7	9	11	13
	3	5	7	9	11	13	15
	5	7	9	11	13	15	17
	7	9	11	13	15	17	19
	9	11	13	15	17	19	21
	11	13	15	17	19	21	23

Kerwin Springer

(ii) State the probability that the sum of the scores from throwing the dice at (b)(i) is

- Prime [1]

From the possibility space table, the prime numbers are:

2, 3, 5, 7, 11, 13, 17, 19, 23.

In total, there were 25 prime numbers out of a total of 36 outcomes.

$$P(the\ sum\ is\ prime) = \frac{Number\ of\ prime\ sums}{Total\ number\ of\ sums}$$

$$= \frac{25}{36}$$

∴ The probability that the sum of the scores is prime is $\frac{25}{36}$.

- Even [1]

There are no even numbers in the possibility space table.

$$P(the\ sum\ is\ even) = \frac{Number\ of\ even\ sums}{Total\ number\ of\ sums}$$

$$= \frac{0}{36}$$

$$= 0$$

Kerwin Springer

∴ The probability that the sum of the scores is even is 0.

(c) Calculate the interquartile range of the data set represented in the stem-and-leaf diagram below. [3]

Stem	Leaf					
3	0	1	3	3		
4	2	4	5			
5	1	6	6	8	9	9
6	4	7	7			
7	3	6				
8	5					
9	2					

Key: 3|0 means 30

The data set has 20 values in total so $n = 20$.

$Q_1 = \frac{1}{4}(n+1)^{th}$ value

$= \frac{1}{4}(20+1)^{th}$

$= \frac{1}{4}(21)^{th}$

$= 5.25^{th}$ value

Looking at the stem-and-leaf diagram, the 5th value is 42 and the 6th value is 44.

Therefore,

$$Q_1 = \frac{42+44}{2}$$

$= 43$

Kerwin Springer

$Q_3 = \frac{3}{4}(n+1)^{th}$ value

$= \frac{3}{4}(20+1)^{th}$

$= \frac{3}{4}(21)^{th}$

$= 15.75^{th}$ value

Looking at the stem-and-leaf diagram, the 15th value is 67 and the 16th value is 67.

Therefore,

$$Q_3 = \frac{67+67}{2}$$

$$= \frac{134}{2}$$

$$= 67$$

Interquartile range, IQR $= Q_3 - Q_1$

$= 67 - 43$

$= 24$

∴ The interquartile range is 24.

Kerwin Springer

(d) Figure 3 shows a box-and-whisker plot which represents a set of numerical data.

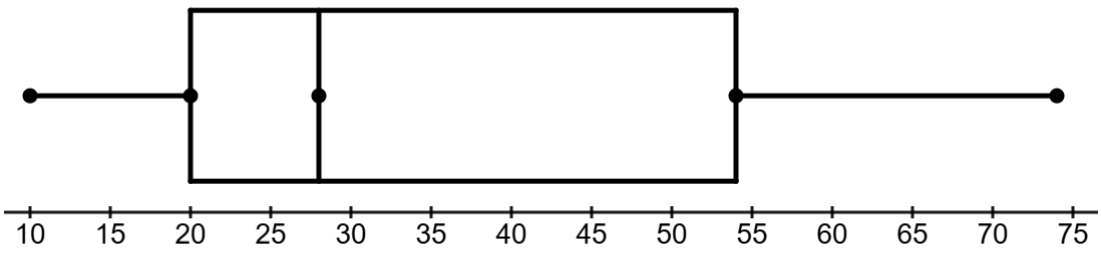


Figure 3. Box-and-whisker plot

(i) State the median of the data set. [1]

Reading off the box-and-whisker plot,

Median = 28.

(ii) Determine the range of the data set. [1]

Range = Highest value − Lowest value

= 74 − 10

= 64

(iii) Describe the shape of the distribution. [1]

Since Q_2 is closer to Q_1, the distribution is positively skewed and since the right whisker is longer, we can say that there is more variation in the upper half of the data.

Best Online Lessons in the Caribbean
WhatsApp **+1868-310-1306** for more information

(e) Five out of 30 students in a class were randomly selected and their heights recorded in centimetres as shown below.

Student	Height
A	135
B	144
C	137
D	146
E	153

Calculate the variance of their heights, in square centimetres. [4]

$$The\ mean\ height,\ \ \bar{x} = \frac{\sum x}{n}$$

$$= \frac{715}{5}$$

$$= 143$$

Student	Height, x	$(x - \bar{x})^2$
A	135	64
B	144	1
C	137	36
D	146	9
E	153	100

Kerwin Springer

$$\sum(x-\bar{x})^2 = 210$$

Variance, $s = \frac{\sum(x-\bar{x})^2}{n-1}$

$= \frac{210}{5-1}$

$= \frac{210}{4}$

$= 52.5\ cm^2$

Total: 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

Kerwin Springer